



**A single dominant axis of locomotor and sensory divergence
across alternative mating strategies in Wellington tree wētā**

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Abstract

Alternative mating strategies can place different demands on movement and mate detection, potentially favouring different combinations of locomotor and sensory traits. In Wellington tree wētā (*Hemideina crassidens*), I asked whether searching males have relatively longer legs and larger eyes, or whether locomotor and sensory dimensions increase alongside weapon size. Using field-collected adults, I compared body-size-adjusted leg lengths, eye length and tympanic organ area among females and three male morphs that mature at the eighth, ninth or tenth instar. Contrary to the searching prediction, leg and sensory dimensions generally increased from eighth-instar males to the heavily armed tenth-instar morph. Most of this divergence followed a common gradient: one axis captured 88.8% of adjusted group-effect variation, although smaller departures from that axis were detectable. After body-size adjustment, correlations among the five locomotor and sensory traits were positive within every group, and larger heads were generally associated with longer legs. Tenth-instar males showed greater relative enlargement of the anterior legs than females, which mature at the same instar. Together, these findings reveal a largely coordinated pattern of locomotor and sensory divergence across alternative mating strategies in this sample. The underlying processes remain unresolved, however, since these adult measurements cannot separate the effects of weaponry from other developmental and sex differences, and positive trait associations may conceal allocation costs when individuals differ in resource acquisition.

Keywords: alternative mating strategies; allometry; morphological integration; phenotypic covariance; sexual selection; trait compensation; wētā

Introduction

Alternative mating strategies allow males to achieve reproductive success through different routes to mate competition. Large or well-armed males may defend females or reproductive resources, whereas smaller males obtain matings through sneaking, satellite behaviour or active searching (Gross, 1996; Shuster & Wade, 2003; Oliveira *et al.*, 2008). These strategies can involve coordinated differences in morphology, physiology, behaviour and life history (West-Eberhard, 2003). A central question is whether their contrasting demands favour distinct trait combinations or whether several functional systems diverge together along a common morphological gradient.

Exaggerated sexual structures can impose developmental or performance costs. Secondary sexual trait compensation occurs when enlarged supporting structures mitigate those costs (Tomkins *et al.*, 2005; Husak & Swallow, 2011). Larger weapons might therefore be accompanied by larger structures that support or manoeuvre the body during contests. Alternatively, structures competing for limited developmental resources may trade off (Nijhout & Emlen, 1998; Emlen, 2001; Moczek & Nijhout, 2004). Variation in resource acquisition can nevertheless elevate several costly traits together, masking allocation costs in comparisons among individuals (Van Noordwijk & de Jong, 1986; Bonduriansky & Day, 2003). Developmental switches may also alter trait covariance among morphs, creating different integration patterns within the same body plan (Snell-Rood *et al.*, 2010; Rohner *et al.*, 2020).

Studies of weapon-bearing insects illustrate this variation. Horn length covaries positively with leg morphology in dung beetles, sometimes more strongly in major than minor males (Tomkins *et al.*, 2005; Rohner *et al.*, 2020). Relatively long-horned rhinoceros beetles also have relatively long wings and foretibiae, but horn size is unrelated to eye size after body-size adjustment (McCullough & Emlen, 2013). In the stag beetle *Prosopocoilus inclinatus*, weapon size is integrated with eye size in major males and with foreleg length in minor males (Ito *et al.*, 2017). Thus, weapon enlargement need not be accompanied by the same supporting traits in every morph.

Locomotor and sensory demands may also differ between males that search for mates and those that defend them. Searching males could benefit from structures that facilitate movement and detection, whereas defenders could benefit from appendages that provide support during combat. In the desert bee *Centris pallida*, small males rely primarily on vision for mate detection and have relatively larger eyes than large males, which rely more on chemosensory cues (Barrett *et al.*, 2021; Barrett & O'Donnell, 2026). Defending males may nevertheless require substantial sensory investment to assess rivals or mating sites. Comparing locomotor and sensory traits together can reveal whether morphological divergence follows these different functional expectations.

Coordinated divergence offers another possibility: several traits may increase together across morphs after adjustment for structural body size, with most differences following one common gradient. The traits need not change by identical percentages, and smaller trait-specific departures may remain. Shared developmental influences may explain this pattern (Rohner *et al.*, 2020). Assessing it requires distinguishing differences among morph means from covariance among

individuals within each morph; coordination at one level does not establish coordination at the other.

The Wellington tree wētā, *Hemideina crassidens*, is a flightless, nocturnal orthopteran in which females shelter communally in tree cavities, hereafter galleries. Males use enlarged heads and mandibles in contests for female-occupied galleries; larger weapons generally confer an advantage in escalated fights (Kelly, 2005, 2006; Kelly & Adams, 2010). Males express three genetically polymorphic, irreversible morphotypes through maturation at the eighth, ninth or tenth juvenile instar (Nason & Kelly, 2020; Kelly, 2024). Tenth-instar males have the most exaggerated weapons and commonly defend galleries. Eighth-instar males may circumvent defenders or search vegetation for undefended females; ninth-instar males combine fighting and sneaking according to social context (Wilson & Kelly, 2019; Kelly, 2025). Similar average reproductive fitness among strategies supports maintenance of the polymorphism (Nason & Kelly, 2020).

I compared the relative expression and allometry of fore-, mid- and hindleg lengths, eye length and tympanic organ area among these male morphs and females. Tympanal membranes receive airborne sound in *H. crassidens*, and the frequency of greatest auditory sensitivity overlaps the spectral peak of conspecific stridulation (Field *et al.*, 1980). Females provide a reference sharing the general body plan without the exaggerated male weapons. I also examined anterior–posterior leg proportions and associations of leg and sensory dimensions with composite head size, used as a proxy for weapon investment.

Three complementary hypotheses guided these comparisons. The searching-and-detection hypothesis predicts relatively longer legs and larger eyes in eighth- than tenth-instar males, assuming these dimensions improve movement and visual detection during searching. Extending this prediction to tympanic organs additionally assumes that searching increases reliance on acoustic cues and that membrane area reflects the relevant performance.

The positive-integration hypothesis predicts relatively larger supporting structures in heavily armed tenth-instar males and positive within-morph weapon–trait associations. Stronger associations in that morph are a further prediction if greater weapon-support demands couple trait expression more tightly. This rationale is most direct for legs; extending it to sensory traits requires a contribution to weapon-associated behaviour. Functional compensation may explain positive integration.

The coordinated-divergence hypothesis predicts that adjusted group differences in locomotor and sensory dimensions largely follow one common gradient. I assessed this prediction through exploratory multivariate analyses, then examined within-group associations and differences among covariance and correlation matrices. The hypotheses address different aspects of morphology and are not mutually exclusive. Finally, because females and tenth-instar males mature at the same instar, I compared their differences with the within-male instar gradient. This separates two descriptive patterns of divergence while retaining the distinction between sex-associated differences and causal effects of weaponry.

Materials and methods

Study site and specimen collection

The sample comprised 313 adult *H. crassidens* collected haphazardly at night from vegetation on Te Pākeka/Maud Island, New Zealand (41°02' S, 173°54' E), during March and April 2016. The island is a 309-ha scientific reserve in the Marlborough Sounds, free of introduced mammalian predators. Vegetation sampling may under-represent gallery occupants, and the analysed male morph composition differed from published reference samples. I did not retain collection microhabitat and search effort in sufficient detail to correct this potential selection bias. The results therefore describe the sampled adults; fixed adult morphology does not guarantee that either group means or within-group covariance represent the population.

Adult males were assigned to eighth-, ninth- or tenth-instar morphotypes using the head-length cut-offs for the 2016 sample (Kelly, 2026). The category boundaries were 18.50579 and 24.15225 mm: males below the lower boundary were assigned to the eighth instar, those above the upper boundary to the tenth instar, and those between or at the boundaries to the ninth instar. I retained these categorisations for all analyses; we did not refit mixture models to this sample. Individuals were euthanized by freezing at −20 °C. The head, pronotum and both legs of each pair were dissected and stored together in a labelled tube preserving the individual's identifier.

Institutional animal-ethics approval was not required for this invertebrate study. The Department of Conservation provided permits for specimen collection and access to Te Pākeka/Maud Island.

Imaging and morphological measurements

Each dissected structure was photographed with a Canon EOS 5D Mark IV camera, Canon EF 100 mm f/2.8L IS USM Macro lens and Canon MT-26EX-RT flashes. The camera was mounted approximately 40 cm above the specimen with its sensor parallel to the imaging surface. A ruler and specimen identifier were photographed in the same plane; images were calibrated from the ruler and given a 5-mm scale bar using ImageJ.

I made measurements in Fiji (Schindelin *et al.*, 2012). Linear measurements used consistent anatomical landmarks and the straight-line tool; tympanic organ area was obtained by tracing the membrane perimeter (Figure 1). Traits were pronotum length, head length and width, eye length, and femur and tibia lengths for all three leg pairs. I measured bilateral structures separately and averaged them; when I measured only one side, I used the available side. Total leg length was the sum of the corresponding femur and tibia lengths.

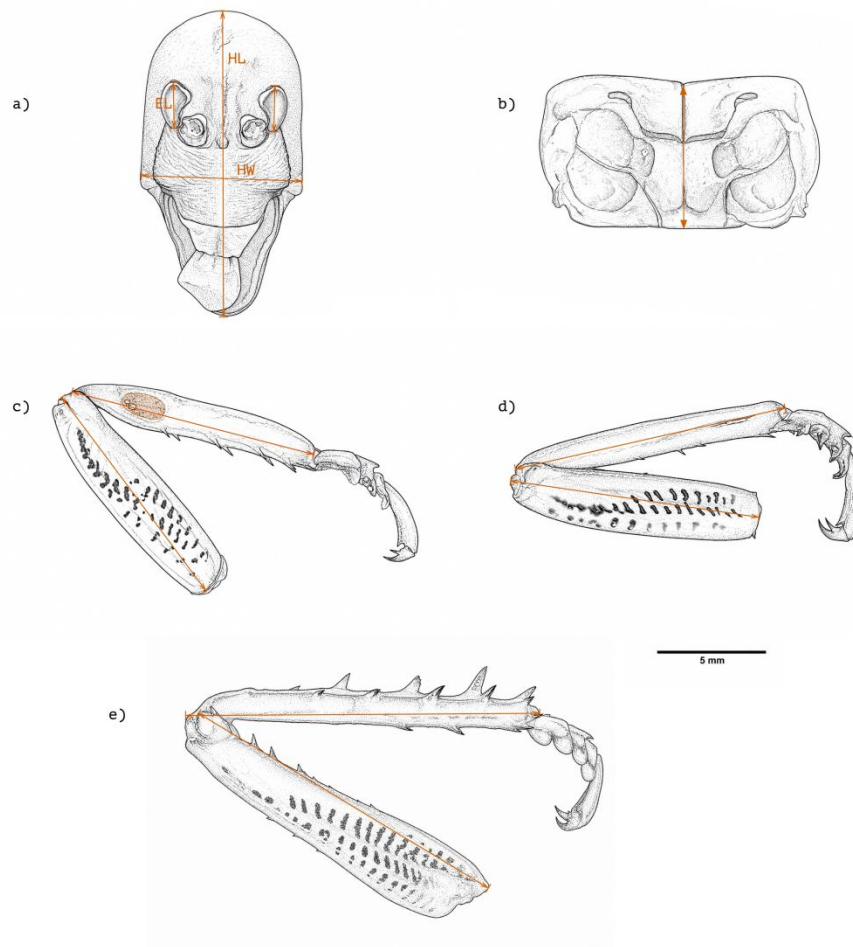


Figure 1: Morphological landmarks used to measure Wellington tree wētā. (a) Dorsal head view showing head length (HL), head width (HW), and eye length (EL). (b) Dorsal pronotum view showing pronotum length. (c–e) Foreleg, midleg, and hindleg, respectively, showing femur and tibia lengths; panel c also indicates the tympanal membrane, whose area was measured by tracing its perimeter. Orange arrows indicate measurement endpoints. Scale bar = 5 mm.

Measurement repeatability

I remeasured 48 individuals twice from the original images, yielding three measurements per trait. The sample contained 12 individuals from each sex or morph group, one from each of 12 recorded size strata. Image order was randomized independently for each round, and previous measurements were not consulted.

For each trait, a random-intercept model estimated repeatability as $R = V_{\text{individual}} / (V_{\text{individual}} + V_{\text{residual}})$. Morph-adjusted and measurement-round-adjusted models, absolute-agreement coefficients, measurement error, round-pair bias and order diagnostics are reported in the supplement (Table S1).

Statistical analysis

Analyses used R 4.4.2 and 4.3.2 (R Core Team, 2025), with *tidyverse* for preparation and visualization (Wickham *et al.*, 2019), base R for ordinary least-squares models, *nlme* for generalized least squares (Pinheiro *et al.*, 2022), and *emmeans* for adjusted means, trends and contrasts (Lenth *et al.*, 2019). Diagnostics used *performance*, *see* and *broom* (Robinson, 2014; Lüdecke *et al.*, 2021a; b). Females and the three male morphs formed the four levels of group.

Pronotum length represented structural body size. Composite head size was the geometric mean of head length and width. Tympanic organ area was square-root transformed to a linear dimension, and all continuous morphological variables were then natural-log transformed. Log-log slopes are consequently allometric exponents. Log pronotum length was centred at 7.2 mm, $\log P_c = \log(\text{pronotum}) - \log(7.2)$, within the body-size range shared by all groups. Adjusted means at $\log P_c = 0$ thus represent interpolation. Data were checked for duplicate identifiers,

unexpected labels, missing or nonpositive measurements and implausible ranges; missing observations were excluded per model.

Dimensionality and covariance

The primary multivariate trait set comprised foreleg, midleg and hindleg length, linearised tympanic organ size and eye length. Head size was excluded because head length contributes to morph assignment. These analyses used the 311 individuals complete for all five traits. The dimensionality and matrix tests are exploratory descriptions of integration and are distinct from the planned marginal contrasts below.

A multivariate model with group-specific body-size slopes estimated adjusted group differences at 7.2 mm. Each log trait was divided by its residual standard deviation from a pooled regression on log pronotum length, and the three male groups were expressed as contrasts with a shared female reference. A design-weighted singular value decomposition quantified the fraction of group-effect variation captured by a rank-one (single-axis) approximation. A test based on 9999 randomizations permuted multivariate residual vectors around the fitted single-axis model and assessed departures from a single axis. This procedure preserves the shared female reference and does not treat repeated copies of females as independent groups. Sensitivity analyses added head size and repeated the six-trait analysis on unstandardized log measurements (Table S7).

Principal component analysis described individual-level size-adjusted variation. For each trait, the adjusted value was its group-specific fitted value at 7.2 mm plus its residual, retaining adjusted group differences while removing the estimated within-group body-size slope. Parallel analysis compared observed correlation-matrix eigenvalues with the 95th percentiles obtained by

177 independently permuting trait columns in 9999 randomizations. This individual-level ordination
178 and the single-axis analysis of group effects address different sources of variation.

179 I calculated within-group covariance and correlation matrices from residuals of group-specific
180 trait–body-size regressions. Omnibus and pairwise Frobenius-distance tests used 9999
181 randomizations of residual-vector group membership, retaining observed group sizes. Pairwise p-
182 values were adjusted by Holm and Benjamini–Hochberg procedures. Adding head size tested
183 sensitivity to trait selection; matrix similarities and rarefaction assessed descriptive structure and
184 unequal sample sizes (Tables S8–S9). I also counted positive and negative within-group pairwise
185 correlations and summarized their range and mean. This sign audit is descriptive: positive
186 phenotypic covariance can mask allocation costs when resource acquisition varies among
187 individuals (Van Noordwijk & de Jong, 1986).

188 **Leg allometry and allocation**

189 Separate leg models compared common and group-specific slopes using nested-model *F*-tests.
190 The principal analysis of relative leg allocation was a repeated-trait generalized least-squares
191 model,

192
$$\log(\text{leg length}) \sim \log P_c \times \text{group} \times \text{leg}.$$

193 Each individual contributed up to three leg measurements. An unstructured residual covariance
194 matrix allowed separate residual variances for the three leg pairs and freely estimated within-
195 individual correlations. It fitted better than heterogeneous compound symmetry (Table S10).
196 Models used restricted maximum likelihood, with type III joint tests and approximate

Satterthwaite denominator degrees of freedom. The three-way interaction tested whether group differences in body-size slope depended on leg identity.

Adjusted means were estimated by leg pair at 7.2 mm and exponentiated to geometric means.

Exponentiated contrasts are ratios, with values above one indicating a larger estimate for the first-named group. Planned comparisons contrasted eighth- with tenth-instar males, ninth-instar males with the eighth/tenth midpoint, and tenth-instar males with females. Difference-in-differences contrasts compared foreleg with hindleg, midleg with hindleg, and the mean of the anterior leg pairs with hindleg. These quantify relative anterior–posterior allocation rather than absolute leg size. Separate femur and tibia models and detailed slope comparisons were exploratory (Tables S14–S16).

Weapon and sensory associations

Weapon–leg integration was tested by adding raw log composite head size to the repeated-trait model,

$$\log L \sim (\log P_c + \log H) \times G \times J,$$

where L is leg length, H is composite head size, G is group and J is leg identity.

The same unstructured residual covariance was retained. Group- and leg-specific partial head-size slopes controlled for pronotum length and were compared within the relevant contrast families. Partial slopes describe weapon–leg associations within groups after adjustment for structural body size.

216 Body-size-adjusted sensory models related log-linearised ear size and log eye length to $\log P_c$,
217 group and their interaction. Common- and group-specific-slope models were compared by nested
218 F -tests, and means were estimated at 7.2 mm. Eye estimates were exponentiated to lengths. Ear
219 estimates were returned to area as $\exp(2\hat{\mu})$, and ear contrast ratios were squared after
220 exponentiation.

221 I tested local associations in one-stage models with group-specific body-size effects plus raw log
222 foretibia length for the ear or raw log head size for the eye. Each anatomical predictor was
223 allowed to interact with group, and a nested-model test compared this with a common partial
224 slope. The corresponding ear model with head size tested weapon–ear covariance. Head size also
225 reflects feeding structures and the substrate bearing the eyes, so these models estimate
226 morphological associations rather than weapon effects alone.

227 **Instar-matched comparisons and precision**

228 The secondary instar comparison used separate group means, body-size slopes and residual
229 variances for each trait:

230
$$\log(\text{trait}) \sim \log P_c \times \text{group}.$$

231 The model was fitted as four independent group regressions, equivalent to a generalized least-
232 squares model with group-specific residual variances. At 7.2 mm pronotum length, I defined the
233 average male change per instar as half the adjusted tenth-minus-eighth log difference, and the
234 instar-matched sex contrast as the adjusted tenth-minus-female log difference. I tested their
235 difference directly, accounting for the shared tenth-instar estimate, with Welch–Satterthwaite
236 degrees of freedom and pointwise 95% confidence intervals. This formulation allows the ninth-

instar mean to depart from the eighth/tenth midpoint; the average male change does not imply equal successive instar increments.

I adopted this specification after diagnostic checks of a more restrictive model with a common body-size slope and equal male instar increments. The original and final estimates are compared in Table S20. The flexible specification was applied to all five traits, without selecting a model by the significance of the biological contrasts. These are separate trait-level comparisons; tests of relative leg allocation came from the repeated-trait model described above. Matching terminal instar does not isolate weapon expression: the sex contrast includes all differences between tenth-instar males and females, and neither contrast identifies a causal developmental effect.

Estimated marginal means and trends are reported with 95% confidence intervals. Planned leg and allocation contrasts and weapon–trait slopes are accompanied by approximate minimum detectable effects at 80% power and $\alpha = 0.05$ for a single two-sided test, calculated from the reported standard errors and t degrees of freedom. For leg contrasts and head–leg slopes, these calculations use the same approximate Satterthwaite degrees of freedom as the corresponding main models (Tables S3, S5 and S19). Exploratory equivalence tests used illustrative regions specified in Tables S5 and S13.

Inference and diagnostics

Holm adjustments were applied within the reported contrast families, including male–female, male-strategy and allocation contrasts, and across the five traits separately for the average male change, matched-sex contrast and their direct difference. The cross-analysis multiplicity audit is a separate sensitivity analysis (Table S12). For the primary contrast families tested with Holm

adjustment, the accompanying simultaneous 95% confidence intervals use Bonferroni adjustment over the same family, as implemented by *emmeans*. Their exclusion of zero or one therefore need not agree with a Holm-adjusted test. Secondary instar-comparison and other 95% intervals are pointwise; the precision tables and minimum detectable effects describe individual comparisons without multiplicity adjustment. Equivalence analyses use pointwise 90% intervals. Exploratory analyses were not treated as additional confirmatory tests.

Graphical diagnostics assessed residual normality, variance and residual–fitted relationships; variance-inflation diagnostics assessed collinearity. Cook’s distance above $4/n$, leverage above $2p/n$, or an absolute standardized residual above 3 flagged observations for inspection, where n is sample size and p is the number of coefficients. These criteria did not trigger automatic deletion.

Results

Measurement repeatability

Measurement repeatability was high across all 11 traits (pooled $R = 0.9763\text{--}0.9997$) and remained high after adjustment for morph ($R = 0.9488\text{--}0.9960$). Differences among measurement rounds were detected for eight traits, but the largest pairwise mean shift was 2.15% of the trait mean (Table S1).

Sample composition and body-size overlap

The analysis included 313 individuals across four groups (Table 1). Pronotum lengths overlapped from 6.99 to 7.49 mm, encompassing the 7.2-mm reference used for adjusted comparisons.

Table 1: Sample sizes and descriptive body-size statistics for females and the three male developmental groups.

Group	Sample size	Pronotum mean	Pronotum SD	Pronotum minimum	Pronotum maximum
Female	118	7.81	0.41	6.86	8.60
Eighth instar	78	6.37	0.37	5.71	7.49
Ninth instar	79	6.93	0.32	6.20	7.73
Tenth instar	38	7.52	0.31	6.99	8.31

278 **Dimensionality and covariance**

279 For the five non-head traits, the dominant axis captured 88.8% of design-weighted group-effect
280 variation (Table 2; Figure 2a). The remaining variation exceeded the single-axis randomization
281 distribution ($p < 0.001$; 9999 randomizations), indicating smaller but detectable departures from a
282 single axis. For all five traits, adjusted male–female differences increased from eighth- through
283 ninth- to tenth-instar males (Figure 2b). Across the 3 trait-set and scaling scenarios, the single-
284 axis share ranged from 88.8% to 98.3% (Table S7).

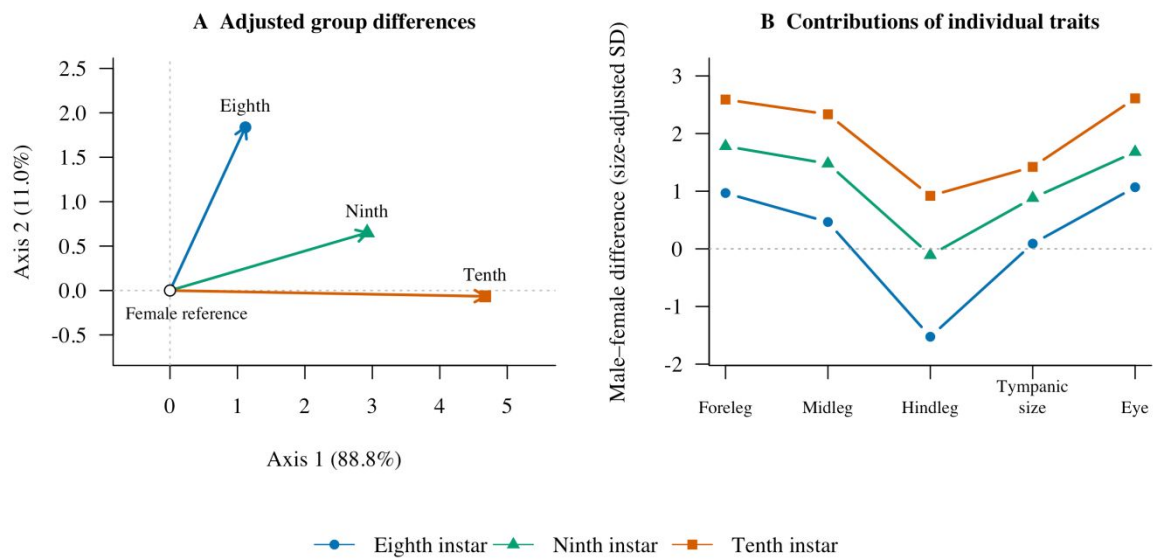


Figure 2: Body-size-adjusted differences between each male morph and females across the five non-head traits. (a) Male–female contrasts projected onto the first two trait axes of the design-weighted decomposition. Percentages give shares of weighted group-effect variation; the female reference is at the origin. (b) Contributions of individual traits, in units of residual standard deviation from pooled trait–body-size regressions. Lines connect traits within each morph for visual comparison. Tympanic size is the square root of membrane area.

285 Parallel analysis of individual-level adjusted measurements retained 1 component, explaining
286 68.7% of variance. The second eigenvalue (0.82) was below its randomization threshold (1.12).
287 All five first-component loadings had the same sign, describing coordinated variation in leg and
288 sensory dimensions. This result concerns individual-level variation and does not negate the
289 smaller departure from a single axis detected among adjusted group means.

290 In the primary five-trait analysis, no difference was detected among groups in covariance
291 matrices ($p = 0.398$) or correlation matrices ($p = 0.848$; 9999 randomizations), and no pairwise
292 comparison survived Holm adjustment (Table S8). When head size was restored in the six-trait
293 sensitivity analysis, a covariance-matrix difference was detected ($p = 0.042$), but a correlation-
294 matrix difference was not ($p = 0.581$). Matrix results thus depended on whether the classification-
295 linked head trait was included.

296 All 40 of the 40 within-group pairwise correlations among the five adjusted traits were positive
297 (Table S18). Mean correlations ranged from 0.45 to 0.52 among groups, with a minimum
298 individual correlation of 0.05. The positive associations spanned all five traits in each group.

Table 2: Structure of morphological integration among the five non-head traits.

Quantity	Value
Single-axis share of group-effect variation	88.8%
Range across analysis scenarios	88.8–98.3%
Residual beyond a single axis (permutation p)	$p < 0.001$
Components retained (parallel analysis)	1
Variance explained by the first component	68.7%
Group difference in covariance matrices (p)	$p = 0.398$
Group difference in correlation matrices (p)	$p = 0.848$
Negative size-adjusted within-group covariances	0 of 40
Traits and groups in the sign audit	5 traits x 4 groups

Locomotor morphology

Relative allocation among leg pairs

Group differences depended strongly on leg identity in the repeated-trait model ($F_{6,301.27} = 71.97, p < 0.001$). Male–female differences therefore varied among forelegs, midlegs and hindlegs (Figure 3; Table S14). At the common body size, forelegs of eighth-, ninth- and tenth-instar males were respectively 5.8% longer (Holm-adjusted $p < 0.001$), 11.4% longer (Holm-adjusted $p < 0.001$) and 16.9% longer (Holm-adjusted $p < 0.001$) than those of females. Corresponding midleg differences were 2.1% longer (Holm-adjusted $p = 0.053$), 7.5% longer (Holm-adjusted $p < 0.001$) and 12.2% longer (Holm-adjusted $p < 0.001$). Hindleg differences were 6.3% shorter (Holm-adjusted $p < 0.001$), 0.5% shorter (Holm-adjusted $p = 0.573$) and 3.9% longer (Holm-adjusted $p < 0.001$).

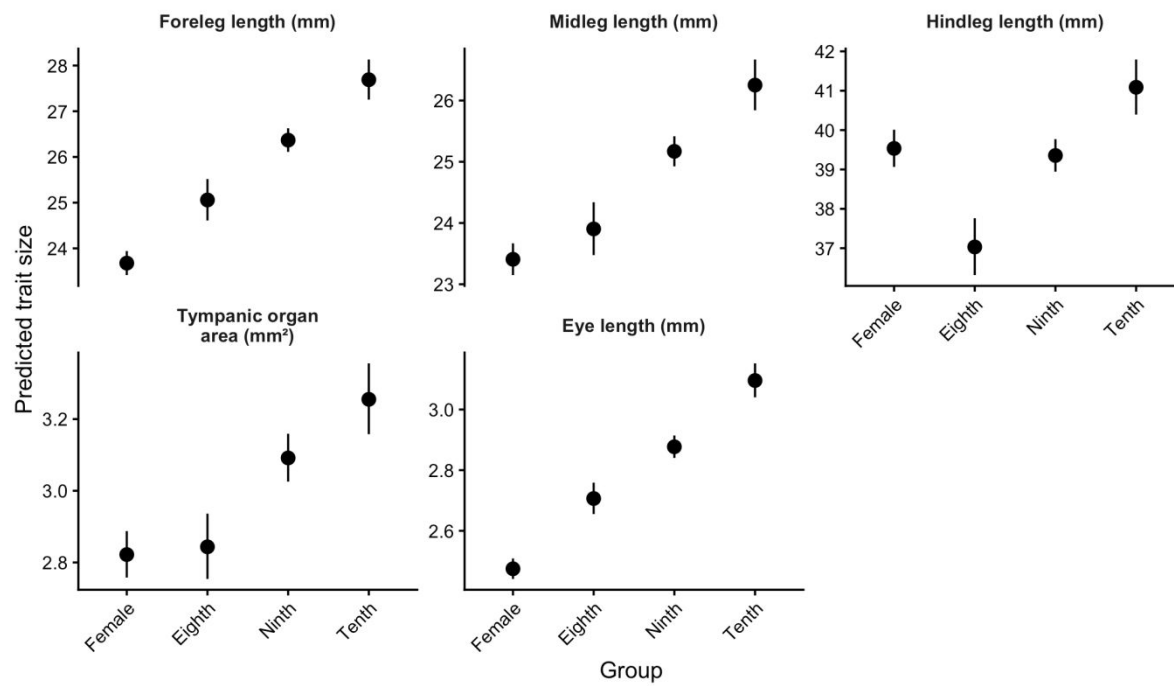


Figure 3: Body-size-adjusted predicted lengths of the three leg pairs and eye, and predicted tympanic organ area, for females and the three male morphotypes at a pronotum length of 7.2 mm. Points are model estimates; bars are 95% confidence intervals. Scales differ among panels, and tympanic organ area is in square millimetres.

310 Tenth-instar males had longer legs than eighth-instar males across all three pairs. Eighth-instar
311 forelegs were 9.5% shorter (Holm-adjusted $p < 0.001$) and hindlegs 9.9% shorter ($p < 0.001$).
312 Ninth-instar estimates were near the eighth/tenth midpoint. Exploratory two-one-sided tests
313 supported equivalence within a $\pm 2.5\%$ region for all three pairs (largest $p = 0.032$; Table S13).
314 The largest absolute midpoint contrast was 0.9%, compared with a maximum minimum-
315 detectable difference of 2.4% (Table S19). The equivalence result is conditional on the chosen
316 region.

317 The planned anterior–posterior allocation contrasts among male morphs were not supported (all
318 Holm-adjusted p -values = 1.000). Tenth-instar males nevertheless showed an anterior shift
319 relative to females: their foreleg-to-hindleg ratio was 12.5% greater (ratio of ratios 1.13; Holm-
320 adjusted $p < 0.001$). All three planned tenth-instar male–female allocation contrasts were
321 supported (largest Holm-adjusted $p < 0.001$).

322 **Allometric variation**

323 All estimated group-by-leg slopes were below one (Figure S4; Table S15). The body-size-by-
324 group interaction was supported ($F_{3,306.04} = 3.53, p = 0.015$), but the body-size-by-group-by-leg
325 interaction was not ($F_{6,301.23} = 1.81, p = 0.096$). Thus, I detected some group variation in
326 scaling without clear evidence that it differed by leg pair. Tenth-instar males had the shallowest
327 point estimates, but corrected comparisons did not establish a consistent morph ordering. The
328 adjusted means best describe group differences; detailed slopes and femur–tibia analyses remain
329 exploratory (Tables S15–S16).

Instar-matched comparisons

The average male change per instar and the matched-sex contrast were positive for all five traits (Table 3; all Holm-adjusted $p < 0.001$).

For forelegs, midlegs, eyes and linearised tympanic organ size, the matched-sex difference exceeded the average male change per instar. Matched male–female differences were 16.9%, 12.2%, 27.0% and 7.8%, compared with average male changes of 5.1%, 4.7%, 7.4% and 3.6% per instar. All four direct comparisons had Holm-adjusted $p < 0.001$.

For hindlegs, the estimated average male change was 5.3% per instar, and the matched-sex difference was 3.9%. Their difference was uncertain (Holm-adjusted $p = 0.158$), so the data did not establish which contrast was larger. The original common-slope model had supported a larger instar change for hindlegs; this conclusion was sensitive to the body-size specification (Table S20).

Table 3: Secondary instar-matched comparisons from models allowing separate group means, body-size slopes and residual variances. Average male change per instar is half the tenth-minus-eighth log contrast; the matched-sex contrast is tenth-instar males minus females. Both are evaluated at 7.2 mm pronotum length. Ear percentages refer to linearised tympanic size. Intervals are pointwise 95% confidence intervals. Difference p-values compare the two log contrasts directly and are Holm-adjusted across five traits.

Trait	n	Mean per instar (%) [95% CI]	Sex contrast (%) [95% CI]	Difference P
Foreleg length	313	5.1 [3.9, 6.4]	16.9 [15.0, 18.9]	<0.001
Midleg length	312	4.7 [3.5, 6.0]	12.2 [10.3, 14.0]	<0.001
Hindleg length	311	5.3 [3.9, 6.6]	3.9 [1.9, 6.0]	0.158
Tympanic organ size	313	3.6 [2.0, 5.3]	7.8 [4.9, 10.7]	<0.001
Eye length	313	7.4 [5.3, 9.6]	27.0 [23.3, 30.8]	<0.001

Weapon–leg integration

After adjustment for pronotum length, 11 of 12 head–leg slopes had 95% confidence intervals entirely above zero; no estimated slope was negative (Table S3). The group-by-leg-by-head interaction was supported ($F_{6,301.6} = 3.62, p = 0.002$), indicating heterogeneity among partial slopes. Planned morph contrasts were nevertheless uniformly nonsignificant after Holm adjustment (Table S4), so they did not establish a simple morph ranking in integration strength. Positive weapon–leg associations were therefore widespread across groups.

Sensory morphology

Adjusted group differences

Group-specific body-size slopes were not supported for ear size ($F_{3,305} = 0.25, p = 0.862$) or eye length ($F_{3,305} = 0.63, p = 0.596$), and adjusted comparisons used common-slope models. At 7.2 mm pronotum length, tenth-instar males had tympanic organ areas 15.3% larger than females (Holm-adjusted $p < 0.001$). Eighth-instar males had areas 12.6% smaller than tenth-instar males ($p < 0.001$).

Tenth-instar males had eyes 25.1% longer than females (Holm-adjusted $p < 0.001$); eighth-instar eyes were 12.6% shorter than tenth-instar eyes ($p < 0.001$). Ninth-instar midpoint contrasts were nonsignificant for both ear ($p = 0.241$) and eye ($p = 0.459$), with exploratory equivalence tests in Table S13. Sensory dimensions followed the same broad eighth-to-tenth-instar gradient as leg length (Figure 3; Table S17).

Local associations and weapon–ear covariance

No group difference in partial slope was detected for ear–foretibia ($F_{3,301} = 1.33, p = 0.264$) or eye–head ($F_{3,301} = 1.97, p = 0.118$) associations. Confidence intervals were entirely positive for 4 of 4 ear–foretibia slopes and 4 of 4 eye–head slopes. Both local anatomical associations were positive across groups. Eye–head slopes ranged from 0.40 to 0.94 and may partly reflect the shared geometry of the head capsule.

The weapon–ear group-by-head interaction was not supported ($F_{3,301} = 1.34, p = 0.261$). The female slope was positive (0.397, 95% confidence interval 0.178–0.615; $p < 0.001$), whereas none of the three male intervals excluded zero. Male estimates ranged from 0.130 to 0.291, and approximate minimum detectable slopes averaged 0.38 and reached 0.64 (Table S5a). No standardized slope met the illustrative ± 0.20 near-zero equivalence criterion (Table S5). The male intervals remain compatible with a range of association strengths, and the interaction did not establish a difference among groups.

Model diagnostics

Residual plots were broadly consistent with the log–log models, with some departures in the tails and at fitted-range extremes. In the secondary instar analysis, permitting group-specific body-size slopes and residual variances addressed discrepancies in predicted leg means and spreads under the constrained model. I retained the flexible group model for these comparisons; we retained the main repeated-trait and sensory models for their respective hypotheses. Diagnostic thresholds did not trigger automatic exclusion of observations.

Discussion

Alternative mating strategies in Wellington tree wētā were associated with coordinated differences across locomotor and sensory morphology. At a common pronotum length, legs, eyes and tympanic organs were generally larger in tenth-instar males than in eighth-instar searching males. Most adjusted group divergence followed one dominant axis, with smaller departures that remained detectable across the sensitivity analyses. These results show how contrasting mating strategies can be associated with a common gradient in several functional systems, even when their behavioural demands suggest different directions of specialization.

The distinction between overall trait enlargement and changes in relative proportions is particularly informative. Male morphs differed mainly in the lengths of all three leg pairs, whereas greater anterior-leg enlargement emerged in comparisons between tenth-instar males and females. The female reference thus revealed a feature of morphological organization that comparisons among male strategies alone would have missed.

Locomotor divergence among strategies

The leg-length gradient ran opposite to the prediction of relatively longer legs in searching males. Ninth-instar males occupied an intermediate position, consistent with their reported flexibility between fighting and sneaking (Wilson & Kelly, 2019; Kelly, 2025), although this position could also reflect maturation at an intermediate instar. Together with the absence of supported anterior–posterior shifts among male morphs, this pattern suggests that the strategies differ more strongly in overall relative leg size than in the distribution of length among leg pairs.

401 The anterior shift in tenth-instar males relative to females has a plausible functional
402 interpretation. Relatively enlarged anterior legs could help heavily armed males stabilize the
403 body, maintain purchase or manoeuvre during contests, as proposed for weapon-associated leg
404 enlargement in beetles (Tomkins *et al.*, 2005; McCullough & Emlen, 2013). Tests of those
405 functions would establish whether the observed proportions improve performance during combat.
406 Group differences in body-size scaling were less consistent than differences in adjusted
407 dimensions, so the evidence for locomotor divergence rests chiefly on relative trait sizes rather
408 than on extrapolating weapon allometry to supporting structures (Kelly, 2026).

409 **Integration within and among morphs**

410 Within-morph associations extended the coordinated pattern beyond differences in group means.
411 Individuals with relatively larger heads generally had relatively longer legs, and all within-group
412 correlations among the five locomotor and sensory traits were positive. Positive weapon–leg
413 relationships resemble those found in male European earwigs, *Forficula auricularia*, and are
414 consistent with compensation if larger supporting structures offset the demands of exaggerated
415 weapons (Tomkins *et al.*, 2005; Husak & Swallow, 2011). These associations were widespread,
416 but planned comparisons did not consistently show stronger integration in the most heavily
417 armed morph.

418 The multivariate results describe substantial integration with scope for more specific differences.
419 Most differences among group averages followed one shared gradient, but smaller departures
420 showed that the traits did not all change together. Within groups, the five locomotor and sensory
421 traits also tended to vary together. I did not detect differences in this pattern of association among
422 groups, although the tests cannot establish that the groups are equivalent. Including head size

changed the covariance result, suggesting that groups may differ in how much individuals vary even when the pattern of associations is similar. Finally, the analyses of individual measurements and group averages answer different questions: a dominant axis among individuals can coexist with smaller additional differences among group averages.

Several processes could produce this organization. Compensatory development, correlated selection and shared developmental influences may promote positive associations. Variation in resource acquisition provides another explanation: individuals with more resources may develop several larger structures even when allocation to one constrains another within a fixed budget (Van Noordwijk & de Jong, 1986). Because I did not measure acquisition and performance, positive covariance cannot distinguish these mechanisms or rule out allocation costs. The present study contributes by establishing the pattern they must explain: broad coordination among traits, accompanied by differences in specific proportions and associations.

Sensory morphology and functional demands

The relatively smaller eyes of eighth-instar males contrast with the enlarged eyes of small searching *Centris pallida* males (Barrett *et al.*, 2021; Barrett & O'Donnell, 2026). Mate searching therefore offers no simple prediction of sensory dimensions across these systems. Defenders may also benefit from sensory investment in assessing rivals or mating sites, while common developmental influences may couple sensory structures to the broader morph gradient. Tympanic organs followed the same general pattern as eyes, although their relationship to searching remains less direct because I did not test reliance on acoustic cues or auditory performance.

444 Comparisons with females suggest a further possibility. All male morphs had relatively larger
445 eyes, and ninth- and tenth-instar males had larger tympanic organs. Some sensory differences
446 may therefore relate to activities shared by males, including mobility or mate searching, rather
447 than to differences among male strategies. These patterns were expressed as shifts in adjusted
448 trait size under common-slope models; sensory exaggeration need not involve steeper allometry,
449 as also reported for crane-fly antennae (Hirani & Bonduriansky, 2026).

450 Local anatomy also shapes interpretation. Eyes share the head capsule with the mandibular
451 apparatus, and tympanic size covaried with foretibia length. Such associations can arise through
452 anatomical integration without demonstrating coordinated sensory and weapon performance.
453 Moreover, ear size increased across male morphs while within-morph head–ear slopes remained
454 imprecise. Coordinated group means can thus coexist with uncertain individual-level
455 associations. Eye length and membrane area are morphological measurements; linking them to
456 acuity, sensitivity or sound detection requires physiological or behavioural evidence.

457 **What the female reference resolves**

458 Females mature at the tenth instar and therefore provide an instar-matched comparison with the
459 most heavily armed males. For anterior legs, eyes and tympanic size, the difference between
460 tenth-instar males and females was larger than the average change per instar across the male
461 morphs. For hindlegs, the two estimates were closer, and their uncertainty prevented a clear
462 ordering. The female comparison therefore reveals substantial sex-associated divergence in
463 anterior-leg and sensory dimensions, while the hindleg result remains less resolved.

The comparison remains descriptive. Female morphology has reproductive, locomotor and sensory demands of its own, and matching instar number neither isolates weapon expression nor controls chronological development time. The estimated differences describe adult group contrasts; they cannot be divided into causal shares attributable to additional moults and weaponry. Longitudinal developmental measurements would show when the different patterns emerge.

Scope and future tests

The genetic basis of morph expression makes these comparisons less dependent on nutritionally induced morph determination than many insect polyphenisms (Kelly, 2024). Environmental and developmental histories may nevertheless affect adult dimensions even at the same pronotum length. Vegetation sampling may also under-represent gallery occupants, and the sampled male composition differed from published references. The findings consequently describe the sampled adults; unequal group sizes, especially the smaller tenth-instar sample, further limit precision for some associations.

The next step is to connect morphological organization to function and development. Tests of whether anterior leg proportions improve contest performance, and whether eye and tympanic dimensions predict sensory performance, would address the functional interpretations directly. Common-garden studies following genetically assigned morphs through successive instars would establish when their differences arise. Resource manipulations could then test whether acquisition masks allocation costs within the coordinated adult phenotype.

Conclusions

Alternative mating strategies in this sample differed along a largely shared gradient in locomotor and sensory dimensions, with additional differences in trait proportions revealed by comparisons with females. Morphological divergence therefore combined broad coordination with more specific departures, rather than following the predicted enlargement of searching-related structures in eighth-instar males. This organization provides a basis for testing how development, resource use and functional demands shape the alternative phenotypes.

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Supplementary material

A single dominant axis of locomotor and sensory divergence across alternative mating strategies in Wellington tree wētā

Overview

This supplement provides measurement-reliability results and detailed weapon–trait and sensory estimates (Tables S1–S6), multivariate and robustness analyses (Tables S7–S13), and additional estimates from the leg and sensory models (Tables S14–S17). Tables S18–S19 summarize within-group correlations and the precision of planned leg contrasts. Figures S1–S4 show within-group correlations, individual-level ordination, body-size distributions and leg allometries.

The distinction between the five-trait reference analysis and head-inclusive sensitivities identifies the trait sets, not their confirmatory status. Dimensionality, matrix comparisons, equivalence-region checks and the cross-analysis multiplicity audit are exploratory. All analyses in the main manuscript and this supplement retain the morph assignments based on the 2016 head-length cut-offs. Reproduction instructions and additional model outputs are supplied with the analysis code.

Supplementary methods

Morph assignment

Male morph assignments used the head-length cut-offs for the 2016 sample (Kelly, 2026): eighth instar, <18.50579 mm; ninth instar, 18.50579–24.15225 mm inclusive; and tenth instar, >24.15225 mm. These categorisations were retained in every analysis. Mixture models were not fitted to this sample.

Measurement reliability

Forty-eight specimens from the 2016 collection were measured on three occasions. The repeatability sample comprised 12 individuals per sex or morph group, with one individual from each of 12 size strata within each group. Repeatability was estimated for pooled, morph-adjusted and morph-plus-round models. Relative technical error of measurement, absolute-agreement ICC(A,1), round-pair bias and measurement-order trends were also calculated. The order analyses were treated as diagnostics; their minimum Benjamini–Hochberg-adjusted p-values were 0.524 for signed drift and 0.867 for absolute-error drift.

Multivariate sensitivity analyses

The reference female-referenced contrast-vector and PCA analyses excluded composite head size because head length contributed to male morph assignment. Each log trait was standardized by its residual standard deviation from a pooled regression on log pronotum length. Sensitivities restored head size and fitted a rank-one (single-axis) approximation to the six unstandardized log traits. Single-axis tests used 9999 residual randomizations; the matrix tests used 9999 randomizations.

Pairwise directional comparisons constrained two male–female contrast vectors to share a direction while leaving the third unconstrained. The angle between each pair was compared with its pair-specific single-axis randomization distribution, with Holm adjustment across the three pairs. Group-stratified bootstrap samples provided percentile 95% intervals for vector angles and differences in magnitude (Table S2).

41 **Matrix comparisons and unequal sample sizes**

42 Within-group covariance and correlation matrices for the five non-head traits were compared
43 using omnibus and pairwise Frobenius-distance randomization tests. Pairwise p-values were
44 adjusted by both Benjamini–Hochberg and Holm procedures. Matrix-element correlations and
45 random-skewers similarities were used as descriptive orientation measures. To evaluate unequal
46 group sizes, each group was repeatedly rarefied to 38 specimens, the size of the tenth-instar male
47 group.

48 **Repeated-leg covariance structure**

49 The main repeated-leg model estimated separate residual variances for fore-, mid- and hindlegs
50 and freely estimated the three within-individual residual correlations. It was compared with
51 compound-symmetry and heterogeneous compound-symmetry structures using fits with identical
52 fixed effects. The selected unstructured model is used for main-text inference and the associated
53 precision calculations.

54 **Secondary integration, equivalence and sampling checks**

55 Main-text weapon–leg relationships were estimated in one stage from raw log composite head
56 size while controlling for pronotum length, group and leg identity. Weapon–ear slopes were
57 likewise estimated in one stage; an exploratory equivalence sensitivity assessed them against an
58 illustrative standardized near-zero region of -0.20 to 0.20 . All-pairs sensory tests were Holm-
59 adjusted within each six-comparison trait family and are separate from the main text’s planned
60 contrast families. For Holm-tested contrasts, simultaneous 95% confidence intervals use
61 Bonferroni adjustment over the stated family; slope and predicted-mean intervals are pointwise.
62 The separate precision estimates use pointwise intervals and approximate minimum detectable

effects for a single two-sided test at 80% power and $\alpha = 0.05$. They do not include a correction for testing several contrasts. Morph-composition χ^2 tests compared the analysed males with two published empirical compositions; they diagnose compositional mismatch but cannot identify its cause because collection microhabitat and gallery-search effort were not retained.

Secondary instar comparisons

The final secondary analysis allowed an independent intercept, body-size slope and residual variance for each group in each trait. Four separate log-trait regressions on centred log pronotum length provide the same estimates and covariance as a fully interacted generalized least-squares model with group-specific residual variances. At 7.2 mm pronotum length, the average male change per instar was $(\mu_{10} - \mu_8)/2$ and the matched-sex contrast was $\mu_{10} - \mu_F$. Their direct difference is $\mu_F - (\mu_8 + \mu_{10})/2$; forming this contrast retains the covariance introduced by the shared tenth-instar estimate. The ninth-instar mean was unconstrained. Contrast variances were sums of squared contrast weights times the group prediction variances, with analytic Welch–Satterthwaite degrees of freedom. Pointwise 95% intervals and three separate five-trait Holm families were used.

The earlier model assumed a common body-size slope, common residual variance and equal male instar increments. Diagnostic checks identified discrepancies in leg means and spreads under that specification. The final flexible model was applied to every trait because the observations represent discrete adult morphs and do not require equal developmental increments. It was not chosen by the significance of the contrasts. Table S20 reports both specifications. These comparisons describe associations among adult groups, not causal effects of additional moults or

84 weaponry. Main-text repeated-leg allocation tests retain their joint covariance model; the
85 secondary instar analysis makes no joint comparison across leg pairs.

86 **Cross-analysis multiplicity audit**

87 The within-family Holm adjustments reported for planned contrasts remain the manuscript's
88 inferential corrections. As a separate sensitivity analysis, ten selected omnibus p-values were
89 adjusted together using Benjamini–Hochberg and Holm procedures. This broader audit is
90 descriptive and does not replace the main-text tests.

91 **Supplementary results**

92 **Reliability and measurement-session effects**

93 All traits were highly repeatable, including after adjustment for morph and measurement round
94 (Table S1). Round effects were detectable for several traits, but their magnitude was small
95 relative to among-individual variation. No evidence of progressive signed or absolute-error drift
96 was detected within remeasurement sessions.

97 **Directional dimensionality**

98 The base five-trait single-axis model captured 88.8% of design-weighted group-effect variation,
99 while the standardized six-trait and raw-log six-trait sensitivities captured 90.3% and 98.3%,
100 respectively. In every case the residual component beyond a single axis exceeded the
101 randomization distribution ($p < 0.001$; Table S7). Parallel analysis retained one component in
102 both the reference five-trait analysis and the six-trait sensitivity (Table S7; Supplementary Figure
103 S2). These exploratory results show one dominant dimension together with a smaller detected
104 secondary direction in adjusted group means.

Matrix structure

The base five-trait omnibus tests detected neither a covariance-matrix difference ($p = 0.398$) nor a correlation-matrix difference ($p = 0.848$; Supplementary Figure S1). The six-trait sensitivity, which restores head size, detected a covariance-matrix difference ($p = 0.042$), but not a correlation-matrix difference ($p = 0.581$). Thus the five-trait null result cannot be generalized to all measured traits; neither nonsignificant test establishes matrix equivalence. No five-trait pairwise matrix comparison survived Holm adjustment (Table S8). Matrix-element correlations, random-skewers similarities and rarefaction summaries are reported in Table S9 rather than interpreted as independent hypothesis tests.

Repeated-leg covariance and secondary analyses

In the alternative covariance-structure fits, unstructured repeated-leg covariance was preferred to heterogeneous compound symmetry (likelihood-ratio statistic = 74.50, $p < 0.001$; Table S10a). Pairwise residual correlations ranged from 0.690 to 0.871 (Table S10b). All 12 one-stage raw-head weapon–leg slope estimates were positive, with 95% confidence intervals excluding zero for 11; planned differences among slopes were imprecise (Tables S3–S4). Weapon–ear slopes did not meet the illustrative equivalence criterion in any group (Table S5).

Sample composition

Male morph composition based on the 2016 categorisations differed from both published empirical references (Table S11a), but the source of the difference cannot be resolved from the archive. The 7.2-mm reference pronotum used for adjusted comparisons fell within the observed range of every group (Supplementary Figure S3).

126 **Sensitivity of the instar comparison**

127 The matched-sex contrast exceeded the average male change per instar for forelegs, midlegs,
128 tympanic size and eyes under both the original and flexible specifications (Table S20). For
129 hindlegs, the original model supported a larger instar change, whereas the flexible model did not
130 distinguish the two contrasts (Holm-adjusted $p = 0.158$). The main text therefore treats the
131 hindleg ordering as uncertain. This result does not demonstrate equality of the contrasts.

132 **Cross-analysis multiplicity**

133 In the selected unstructured model, the group-by-leg allocation effect and the raw-head weapon–
134 leg interaction remained supported in the ten-test audit. The body-size-by-group-by-leg
135 interaction was unsupported before cross-analysis adjustment. The body-size-by-group term was
136 supported within its model but did not survive the broader ten-test audit (Table S12), reinforcing
137 the exploratory status of detailed allometric comparisons.

138

Supplementary tables

Table S1. Repeatability, agreement and measurement-round diagnostics for 11 morphological traits measured three times.

Trait	Pooled	Morph-adjusted	Morph + round	rTEM (%)	ICC(A,1)	Max. bias (%)	Holm p
Pronotum length	0.9950 (0.9916–0.9968)	0.9819 (0.9690–0.9887)	0.9942 (0.9900–0.9965)	0.65	0.9951	1.08	<0.001
Head length	0.9997 (0.9995–0.9998)	0.9960 (0.9931–0.9976)	0.9982 (0.9968–0.9989)	0.48	0.9997	0.70	<0.001
Head width	0.9991 (0.9984–0.9994)	0.9886 (0.9810–0.9932)	0.9937 (0.9894–0.9962)	0.60	0.9991	0.81	<0.001
Forefemur length	0.9938 (0.9893–0.9962)	0.9794 (0.9642–0.9869)	0.9803 (0.9657–0.9876)	0.72	0.9939	0.36	0.136
Foretibia length	0.9934 (0.9887–0.9960)	0.9817 (0.9687–0.9889)	0.9864 (0.9765–0.9917)	0.69	0.9935	0.66	<0.001
Midfemur length	0.9946 (0.9911–0.9967)	0.9824 (0.9707–0.9893)	0.9851 (0.9750–0.9909)	0.61	0.9947	0.44	0.001
Midtibia length	0.9959 (0.9930–0.9975)	0.9880 (0.9790–0.9925)	0.9880 (0.9788–0.9926)	0.54	0.9959	0.16	0.705
Hindfemur length	0.9955 (0.9924–0.9972)	0.9835 (0.9724–0.9898)	0.9847 (0.9742–0.9905)	0.64	0.9955	0.39	0.039
Hindtibia length	0.9868 (0.9773–0.9919)	0.9562 (0.9249–0.9733)	0.9596 (0.9302–0.9754)	1.21	0.9868	0.73	0.039
Ear area	0.9763 (0.9610–0.9855)	0.9488 (0.9158–0.9691)	0.9565 (0.9276–0.9736)	2.57	0.9765	2.15	0.001
Eye length	0.9906 (0.9841–0.9942)	0.9717 (0.9527–0.9826)	0.9714 (0.9519–0.9824)	1.13	0.9906	0.23	0.705

Note. Repeatability values are individual-level ICCs with 95% bootstrap confidence intervals.

rTEM is the within-individual standard deviation divided by the trait mean $\times 100$. ICC(A,1) is the single-measure absolute-agreement coefficient. Maximum bias is the largest absolute round-pair mean difference as a percentage of the trait mean. Round-effect p-values are Holm-adjusted across traits.

Table S2. Exploratory pairwise descriptions of standardized female-referenced contrast vectors for the five nonclassification traits. D8, D9 and D10 denote the female-referenced eighth-, ninth- and tenth-instar male vectors. The global single-axis test is reported separately in Table S7.

Pair	Angle (°)	Lower 95%	Upper 95%	Angle Holm p	Δ magnitude	Δ lower 95%	Δ upper 95%
D8–D9	46.3	25.0	65.5	0.006	0.84	0.15	1.35
D8–D10	59.5	36.9	79.7	0.018	2.52	1.65	3.17
D9–D10	13.6	7.9	22.0	0.003	1.68	1.11	2.22

Table S3. Group- and leg-specific partial slopes of log leg length on log composite head size (the geometric mean of head length and width), controlling for log pronotum length in the repeated-trait model. Confidence intervals and individual slope p-values are unadjusted. Minimum detectable slopes are approximate values for a single two-sided test at 80% power and $\alpha = 0.05$, using the same approximate Satterthwaite degrees of freedom as the slope intervals.

Group	Leg	Slope	Lower 95% CI	Upper 95% CI	p-value	Minimum detectable slope
Female	Foreleg	0.419	0.281	0.557	<0.001	0.20
Eighth instar	Foreleg	0.459	0.347	0.570	<0.001	0.16
Ninth instar	Foreleg	0.323	0.216	0.430	<0.001	0.15
Tenth instar	Foreleg	0.234	-0.050	0.518	0.106	0.41
Female	Midleg	0.422	0.288	0.555	<0.001	0.19
Eighth instar	Midleg	0.472	0.364	0.580	<0.001	0.15
Ninth instar	Midleg	0.361	0.257	0.464	<0.001	0.15
Tenth instar	Midleg	0.367	0.093	0.641	0.009	0.39
Female	Hindleg	0.633	0.489	0.777	<0.001	0.21
Eighth instar	Hindleg	0.348	0.231	0.466	<0.001	0.17
Ninth instar	Hindleg	0.365	0.253	0.477	<0.001	0.16
Tenth instar	Hindleg	0.533	0.237	0.830	<0.001	0.42

Table S4. Planned contrasts among weapon–leg slopes. P-values are Holm-adjusted and simultaneous 95% confidence intervals are Bonferroni-adjusted within the three-contrast family for each leg pair.

Leg	Contrast	Difference	Lower 95% CI	Upper 95% CI	Adjusted p
Foreleg	Eighth instar - Tenth instar	0.224	-0.149	0.598	0.448
Foreleg	Ninth instar - midpoint(Eighth instar,Tenth instar)	-0.024	-0.252	0.204	0.804
Foreleg	Tenth instar - Female	-0.185	-0.571	0.201	0.499
Midleg	Eighth instar - Tenth instar	0.105	-0.256	0.466	1.000
Midleg	Ninth instar - midpoint(Eighth instar,Tenth instar)	-0.059	-0.279	0.161	1.000
Midleg	Tenth instar - Female	-0.055	-0.428	0.319	1.000
Hindleg	Eighth instar - Tenth instar	-0.185	-0.576	0.205	0.762
Hindleg	Ninth instar - midpoint(Eighth instar,Tenth instar)	-0.076	-0.314	0.163	0.891
Hindleg	Tenth instar - Female	-0.099	-0.503	0.304	0.891

Table S5a. Within-group partial slopes of log linearised tympanic size on log composite head size, controlling for log pronotum length. Confidence intervals are pointwise. Minimum detectable slopes are approximate values for a single two-sided test at 80% power and $\alpha = 0.05$.

Group	Slope	SE	Lower 95% CI	Upper 95% CI	Minimum detectable slope
Female	0.397	0.111	0.178	0.615	0.31
Eighth instar	0.167	0.090	-0.010	0.344	0.25
Ninth instar	0.130	0.086	-0.039	0.300	0.24
Tenth instar	0.291	0.228	-0.159	0.740	0.64

Table S5b. Within-group standardized partial slopes of log linearized tympanal size on raw log composite head size, controlling for log pronotum length. Linearized tympanal size is the square root of tympanal area. Equivalence required the complete 90% confidence interval to fall within -0.20 to 0.20 ; robust slopes are Huber M-estimates.

Group	n	Standardized slope	Lower 90% CI	Upper 90% CI	Robust slope	TOST p	Equivalent within ± 0.20
Female	118	0.466	0.255	0.676	0.437	0.981	No
Eighth instar	78	0.275	0.043	0.507	0.263	0.705	No
Ninth instar	79	0.167	-0.033	0.367	0.172	0.392	No
Tenth instar	38	0.218	-0.077	0.513	0.221	0.541	No

Table S6. Exploratory all-pairs group contrasts in size-adjusted tympanal area and eye length at a pronotum length of 7.2 mm. These six-pair families differ from the planned contrast families in the main manuscript. Ratios greater than one indicate that the first named group had the larger predicted trait. P-values are Holm-adjusted and simultaneous 95% confidence intervals are Bonferroni-adjusted across the six group pairs within each trait. Because the adjustments differ, a Holm-adjusted p-value below 0.05 can accompany an interval containing one.

Trait	Contrast	Ratio	Lower 95% CI	Upper 95% CI	Adjusted p
Tympanal area	Female - Eighth instar	0.992	0.930	1.059	0.757
Tympanal area	Female - Ninth instar	0.913	0.870	0.957	<0.001
Tympanal area	Female - Tenth instar	0.867	0.828	0.908	<0.001
Tympanal area	Eighth instar - Ninth instar	0.920	0.879	0.962	<0.001
Tympanal area	Eighth instar - Tenth instar	0.874	0.818	0.933	<0.001
Tympanal area	Ninth instar - Tenth instar	0.950	0.901	1.001	0.020
Eye length	Female - Eighth instar	0.914	0.879	0.951	<0.001
Eye length	Female - Ninth instar	0.860	0.836	0.885	<0.001
Eye length	Female - Tenth instar	0.799	0.778	0.822	<0.001
Eye length	Eighth instar - Ninth instar	0.941	0.916	0.966	<0.001
Eye length	Eighth instar - Tenth instar	0.874	0.841	0.909	<0.001
Eye length	Ninth instar - Tenth instar	0.929	0.901	0.959	<0.001

Table S7. Sensitivity of dimensionality to trait definition and scaling. For single-axis analyses, the null reference is the 95th percentile of the residual share *beyond* a single axis; it is compared with 100% minus the reported single-axis share. PCA rows instead compare an individual-level eigenvalue with its parallel-analysis threshold.

Approach	Scenario	Estimate	Null reference	P-value	Conclusion
Female-referenced single-axis test	Five traits, standardized; head excluded (base scenario)	88.8% captured by a single axis	Residual share: 95th percentile = 2.1%	<0.001	Departure from a single axis detected
Female-referenced single-axis test	Six traits, standardized; head included	90.3% captured by a single axis	Residual share: 95th percentile = 1.8%	<0.001	Departure from a single axis detected
Female-referenced single-axis test	Six traits on raw log scale	98.3% captured by a single axis	Residual share: 95th percentile = 0.4%	<0.001	Departure from a single axis detected
PCA with parallel analysis	Five traits, standardized; head excluded (base scenario)	PC1 = 68.7%; eigenvalue 3.433	95th-percentile eigenvalue = 1.238	—	One component retained
PCA with parallel analysis	Six traits, standardized; head included	PC1 = 68.6%; eigenvalue 4.116	95th-percentile eigenvalue = 1.268	—	One component retained

182 **Table S8a.** Omnibus permutation comparisons of within-group covariance and correlation
183 matrices.

Trait set	Matrix	Statistic	Permutations	p-value
Five non-head traits (base scenario)	Covariance	116.446	9,999	0.398
Five non-head traits (base scenario)	Correlation	0.360	9,999	0.848
Six traits including head (sensitivity)	Covariance	325.208	9,999	0.042
Six traits including head (sensitivity)	Correlation	0.825	9,999	0.581

184 **Table S8b.** Pairwise permutation comparisons of within-group covariance and correlation
185 matrices. P-values are reported with both Benjamini–Hochberg and Holm adjustment across the
186 six group pairs within each matrix type.

Matrix	Pair	Frobenius distance	Raw p	BH p	Holm p
Covariance	Female – Eighth instar	0.989	0.314	0.471	1.000
Covariance	Female – Ninth instar	0.994	0.298	0.471	1.000
Covariance	Female – Tenth instar	0.924	0.505	0.606	1.000
Covariance	Eighth instar – Ninth instar	0.585	0.929	0.929	1.000
Covariance	Eighth instar – Tenth instar	1.546	0.115	0.471	0.692
Covariance	Ninth instar – Tenth instar	1.420	0.199	0.471	0.997
Correlation	Female – Eighth instar	0.278	0.935	0.935	1.000
Correlation	Female – Ninth instar	0.325	0.804	0.935	1.000
Correlation	Female – Tenth instar	0.620	0.434	0.935	1.000
Correlation	Eighth instar – Ninth instar	0.421	0.738	0.935	1.000
Correlation	Eighth instar – Tenth instar	0.598	0.662	0.935	1.000
Correlation	Ninth instar – Tenth instar	0.483	0.694	0.935	1.000

187

Table S9. Descriptive matrix similarities and unequal-sample-size rarefaction. Rarefaction used 2,000 subsamples that reduced each group to 38 specimens; intervals are the 2.5th and 97.5th percentiles of the resulting distance distribution.

Matrix	Pair	Element correlation	Random-skewers similarity	Rarefied distance, median (95% interval)	Target n
Covariance	Female – Eighth instar	0.884	0.969	1.399 (0.765–2.471)	38
Covariance	Female – Ninth instar	0.902	0.975	1.363 (0.720–2.728)	38
Covariance	Female – Tenth instar	0.817	0.955	1.096 (0.657–2.163)	38
Covariance	Eighth instar – Ninth instar	0.956	0.981	1.145 (0.631–2.093)	38
Covariance	Eighth instar – Tenth instar	0.717	0.935	1.669 (0.973–2.502)	38
Covariance	Ninth instar – Tenth instar	0.835	0.965	1.515 (0.787–2.585)	38
Correlation	Female – Eighth instar	0.987	0.993	0.664 (0.327–1.148)	38
Correlation	Female – Ninth instar	0.993	0.994	0.624 (0.289–1.314)	38
Correlation	Female – Tenth instar	0.947	0.972	0.733 (0.454–1.264)	38
Correlation	Eighth instar – Ninth instar	0.984	0.987	0.671 (0.337–1.221)	38
Correlation	Eighth instar – Tenth instar	0.958	0.974	0.703 (0.390–1.139)	38
Correlation	Ninth instar – Tenth instar	0.952	0.981	0.594 (0.354–0.954)	38

192 **Table S10a.** Comparison of covariance structures for the repeated-leg model. All models contain
193 the same fixed-effects specification and were fitted by REML. The selected unstructured fit is
194 used for main-text inference.

Model	Parameters	AIC	BIC	Log likelihood	Comparison	Likelihood-ratio statistic	p-value
Compound symmetry	26	-4,009.84	-3,884.64	2,030.92	—		—
Heterogeneous compound symmetry	28	-4,019.19	-3,884.36	2,037.60	1 vs 2	13.35	0.001
Unstructured covariance	30	-4,089.70	-3,945.23	2,074.85	2 vs 3	74.50	<0.001

195 **Table S10b.** Estimated parameters from the selected unstructured covariance model.

Parameter	Estimate
Foreleg-midleg residual correlation	0.871
Foreleg-hindleg residual correlation	0.690
Midleg-hindleg residual correlation	0.725
Midleg/foreleg residual SD ratio	0.997
Hindleg/foreleg residual SD ratio	1.074

196

Table S11a. Male morph composition under the 2016 categorisations compared with published reference samples. Observed counts are ordered as eighth / ninth / tenth instar.

Reference	Observed	Expected eighth	Expected ninth	Expected tenth	χ^2	df	p-value
Kelly 2026 retained assignments	78 / 79 / 38	44.1%	29.1%	26.8%	13.36	2	0.001
Kelly and Adams 2010 field sample	78 / 79 / 38	43.3%	30.9%	25.8%	9.29	2	0.010

Table S11b. Head-size and group contributions to eye length when pronotum, composite head size and group are fitted jointly.

Analysis	Numerator df	Residual df	F	p-value
Head-size contribution to eye length after pronotum and group	1	307	112.99	<0.001
Group effect on eye length after pronotum and head size	3	307	10.65	<0.001

202 **Table S11c.** Observed composite head-size ranges. The reversed common bounds show that the
203 four groups have no shared head-size support.

Group	n	Minimum head size	Maximum head size	Common lower bound	Common upper bound	Common support
Female	118	8.491	11.246	16.818	11.246	No
Eighth instar	78	9.794	14.034	16.818	11.246	No
Ninth instar	79	13.321	17.299	16.818	11.246	No
Tenth instar	38	16.818	19.673	16.818	11.246	No

204

Table S12. Cross-analysis multiplicity sensitivity for ten selected omnibus tests. The repeated-leg entries use the selected unstructured covariance model and the weapon–leg entry uses raw composite head size, as in the main manuscript. This descriptive audit does not replace the within-family Holm adjustments used for planned contrasts.

Analysis	Tier	Raw p	BH p	Holm p
Leg allocation group x leg	Primary	<0.001	<0.001	<0.001
Leg allometry body size x group	Secondary	0.015	0.051	0.122
Leg allometry body size x group x leg	Secondary	0.096	0.236	0.671
Raw head size x group x leg	Primary	0.002	0.009	0.016
Ear body-size slope x group	Secondary	0.862	0.862	1.000
Eye body-size slope x group	Secondary	0.596	0.745	1.000
Raw ear-foretibia slope x group	Secondary	0.264	0.377	1.000
Raw eye-head slope x group	Secondary	0.118	0.236	0.707
Raw weapon-ear slope x group	Secondary	0.261	0.377	1.000
Five-trait correlation-matrix equality	Exploratory	0.848	0.862	1.000

Table S13. Exploratory sensitivity of ninth-instar midpoint equivalence to illustrative $\pm 2.5\%$ and $\pm 5\%$ regions. Equivalence under each stated region requires the complete 90% confidence interval to fall within it; lack of equivalence is not evidence of a difference.

Trait	Scale	Equivalence region	Ratio	Lower 90% CI	Upper 90% CI	TOST p	Equivalent
Foreleg	Length ratio	$\pm 2.5\%$	1.001	0.988	1.014	0.001	Yes
Midleg	Length ratio	$\pm 2.5\%$	1.005	0.992	1.018	0.006	Yes
Hindleg	Length ratio	$\pm 2.5\%$	1.009	0.995	1.023	0.032	Yes
Foreleg	Length ratio	$\pm 5\%$	1.001	0.988	1.014	<0.001	Yes
Midleg	Length ratio	$\pm 5\%$	1.005	0.992	1.018	<0.001	Yes
Hindleg	Length ratio	$\pm 5\%$	1.009	0.995	1.023	<0.001	Yes
Tympanal area	area ratio	$\pm 2.5\%$	1.016	0.994	1.039	0.263	No
Tympanal area	area ratio	$\pm 5\%$	1.016	0.994	1.039	0.009	Yes
Eye length	length ratio	$\pm 2.5\%$	0.994	0.981	1.007	0.009	Yes
Eye length	length ratio	$\pm 5\%$	0.994	0.981	1.007	<0.001	Yes

Detailed estimates from the main-text models

The following tables report detailed estimates from the models presented in the main manuscript, including estimates underlying its figures and selected contrasts. They are presented separately from the exploratory and sensitivity analyses in Tables S1–S13.

Table S14. Male–female contrasts in predicted leg length at a pronotum length of 7.2 mm from the main-text repeated-trait model. Percent differences are relative to females; positive values indicate longer legs in males. Confidence limits are for the male-to-female ratio. P-values are Holm-adjusted and simultaneous 95% confidence limits are Bonferroni-adjusted across the three male–female contrasts within each leg pair.

Leg	Contrast	Difference (%)	Ratio lower 95% CI	Ratio upper 95% CI	Adjusted p
Foreleg	Eighth instar - Female	5.8	1.031	1.086	<0.001
Foreleg	Ninth instar - Female	11.4	1.094	1.134	<0.001
Foreleg	Tenth instar - Female	16.9	1.142	1.198	<0.001
Midleg	Eighth instar - Female	2.1	0.995	1.048	0.053
Midleg	Ninth instar - Female	7.5	1.056	1.095	<0.001
Midleg	Tenth instar - Female	12.2	1.095	1.148	<0.001
Hindleg	Eighth instar - Female	-6.3	0.911	0.963	<0.001
Hindleg	Ninth instar - Female	-0.5	0.976	1.015	0.573
Hindleg	Tenth instar - Female	3.9	1.013	1.066	<0.001

224 **Table S15.** Group- and leg-specific allometric slopes of log leg length on log pronotum length
225 from the main-text repeated-trait model. Confidence intervals describe individual slopes. Because
226 the omnibus group-by-leg slope interaction was not supported, these estimates are descriptive
227 rather than evidence of morph-specific slope differences.

Group	Leg	Slope	Lower 95% CI	Upper 95% CI
Female	Foreleg	0.711	0.595	0.827
Eighth instar	Foreleg	0.741	0.609	0.873
Ninth instar	Foreleg	0.604	0.441	0.766
Tenth instar	Foreleg	0.332	0.062	0.601
Female	Midleg	0.727	0.612	0.843
Eighth instar	Midleg	0.690	0.558	0.822
Ninth instar	Midleg	0.655	0.492	0.817
Tenth instar	Midleg	0.329	0.060	0.597
Female	Hindleg	0.809	0.685	0.934
Eighth instar	Hindleg	0.646	0.504	0.789
Ninth instar	Hindleg	0.569	0.395	0.744
Tenth instar	Hindleg	0.273	-0.017	0.562

Table S16. Exploratory nested-model tests for group-specific allometric slopes of femur and tibia segments. These are six separate ordinary least-squares comparisons, not contrasts from the joint repeated-trait model; p-values are unadjusted.

Segment	Numerator df	Residual df	F	p-value
Forefemur: logP_c x group	3	305	4.06	0.007
Foretibia: logP_c x group	3	305	1.45	0.228
Midfemur: logP_c x group	3	304	2.82	0.039
Midtibia: logP_c x group	3	304	1.95	0.121
Hindfemur: logP_c x group	3	304	3.79	0.011
Hindtibia: logP_c x group	3	303	4.20	0.006

Table S17. Body-size-adjusted predicted tympanic organ area and eye length at a pronotum length of 7.2 mm from the main-text sensory models. Estimates and 95% confidence intervals are back-transformed to the stated units; these are group predictions, not pairwise contrasts.

Trait	Group	Estimate	Lower 95% CI	Upper 95% CI
Tympanic organ area (mm ²)	Female	2.822	2.758	2.888
Tympanic organ area (mm ²)	Eighth instar	2.844	2.754	2.936
Tympanic organ area (mm ²)	Ninth instar	3.092	3.025	3.159
Tympanic organ area (mm ²)	Tenth instar	3.255	3.158	3.355
Eye length (mm)	Female	2.475	2.441	2.509
Eye length (mm)	Eighth instar	2.707	2.655	2.759
Eye length (mm)	Ninth instar	2.877	2.840	2.914
Eye length (mm)	Tenth instar	3.096	3.040	3.152

Table S18. Summary of within-group correlations among the five size-adjusted traits. This descriptive sign audit identifies observed negative phenotypic associations; it cannot establish the presence or absence of an underlying resource-allocation cost.

Group	Pairwise correlations	Positive	Negative	Mean <i>r</i>	Minimum <i>r</i>	Maximum <i>r</i>
Eighth instar	10	10	0	0.45	0.10	0.90
Female	10	10	0	0.47	0.05	0.84
Ninth instar	10	10	0	0.52	0.16	0.88
Tenth instar	10	10	0	0.52	0.35	0.86

Note. All observed size-adjusted within-group correlations in the primary five-trait set were positive. No negative phenotypic association was detected, but this does not demonstrate the absence of an allocation cost, because variation among individuals in resource acquisition generates positive covariance among costly traits even when those traits compete for allocation (Van Noordwijk & de Jong, 1986).

246 **Table S19.** Precision of planned body-size-adjusted leg contrasts. Pointwise 95% intervals and
247 approximate minimum detectable effects use the main model’s standard errors and Satterthwaite
248 degrees of freedom. Detectable effects assume 80% power for a single two-sided test at $\alpha = 0.05$,
249 without multiplicity adjustment. Allocation contrasts compare leg pairs and are expressed as
250 ratios of ratios. These precision estimates do not themselves establish equivalence (Table S13).

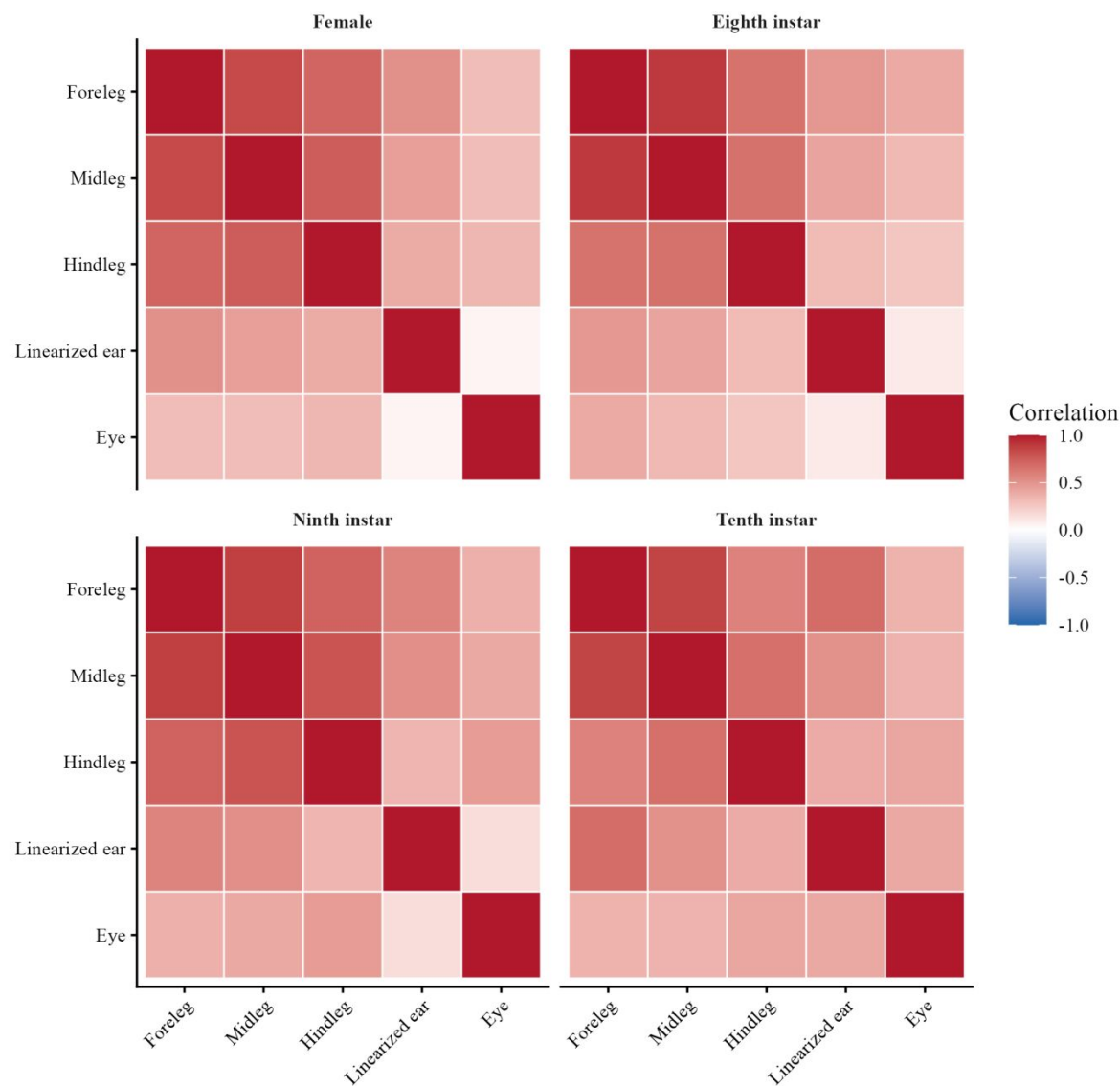
Leg	Contrast	Difference (%)	Ratio lower 95%	Ratio upper 95%	Minimum detectable (%)
Foreleg	Tenth instar - Eighth instar	10.5	1.079	1.132	3.5
Foreleg	Ninth instar - midpoint(Eighth instar, Tenth instar)	0.1	0.986	1.017	2.2
Foreleg	Tenth instar - Female	16.9	1.147	1.192	2.8
Midleg	Tenth instar - Eighth instar	9.8	1.072	1.125	3.5
Midleg	Ninth instar - midpoint(Eighth instar, Tenth instar)	0.5	0.989	1.020	2.2
Midleg	Tenth instar - Female	12.2	1.100	1.143	2.8
Hindleg	Tenth instar - Eighth instar	11.0	1.081	1.139	3.8
Hindleg	Ninth instar - midpoint(Eighth instar, Tenth instar)	0.9	0.992	1.026	2.4
Hindleg	Tenth instar - Female	3.9	1.018	1.061	3.0
All legs (allocation)	Tenth instar - Eighth instar: foreleg relative to hindleg	-0.4	0.976	1.016	2.9
All legs (allocation)	Tenth instar - Female: foreleg relative to hindleg	12.5	1.108	1.143	2.3
All legs (allocation)	Tenth instar - Female: midleg relative to hindleg	7.9	1.063	1.095	2.2

Table S20. Sensitivity of the secondary instar comparison to model specification. The original model imposed a common body-size slope, common residual variance and equal male instar increments. The final model allows separate group means, body-size slopes and residual variances. Average male change per instar and the matched-sex contrast are evaluated at 7.2 mm pronotum length and expressed as percentages of length or linearised tympanic size. The difference interval is pointwise and on the log scale; its p-value is Holm-adjusted across five traits within each specification. Positive differences favour the average male change; negative differences favour the matched-sex contrast.

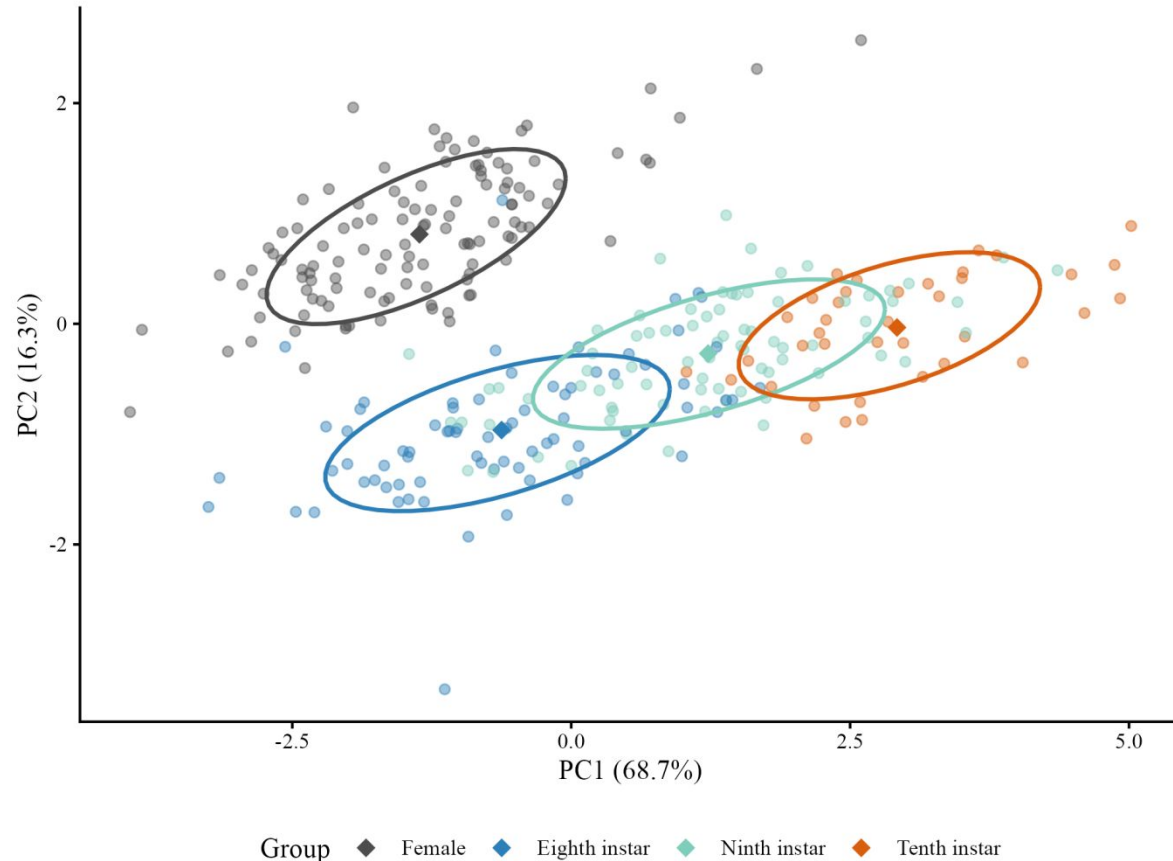
Trait	Model	Mean per instar (%)	Sex contrast (%)	Difference 95% CI (log)	Difference Holm p
Foreleg length	Original	5.05	15.92	-0.1106 to -0.0864	<0.001
Foreleg length	Final	5.12	16.95	-0.1224 to -0.0908	<0.001
Midleg length	Original	4.30	10.91	-0.0734 to -0.0494	<0.001
Midleg length	Final	4.73	12.15	-0.0842 to -0.0527	<0.001
Hindleg length	Original	4.55	2.22	0.0093 to 0.0357	<0.001
Hindleg length	Final	5.25	3.93	-0.0050 to 0.0303	0.158
Tympanic organ size	Original	3.55	7.91	-0.0572 to -0.0252	<0.001
Tympanic organ size	Final	3.61	7.77	-0.0611 to -0.0177	<0.001
Eye length	Original	6.86	24.64	-0.1730 to -0.1349	<0.001
Eye length	Final	7.44	27.01	-0.1929 to -0.1419	<0.001

Supplementary figures

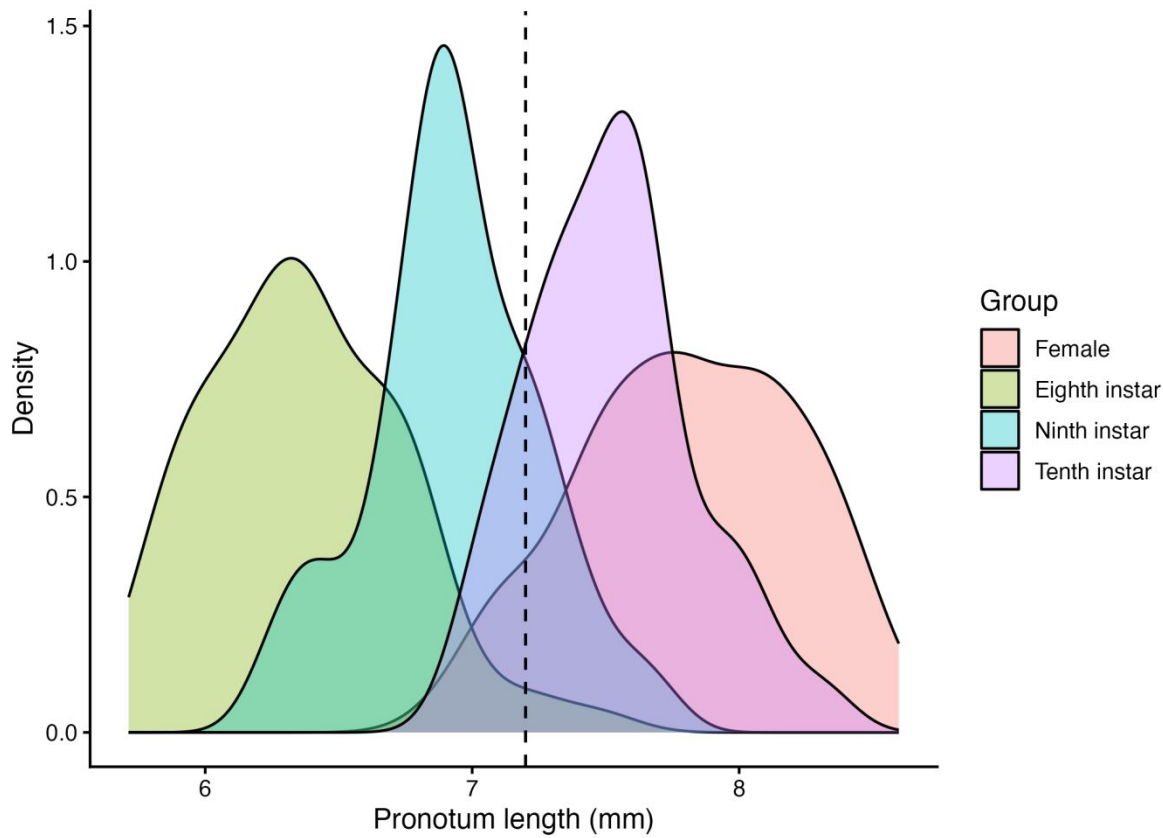
Supplementary Figure S1. Within-group correlation matrices of size-adjusted residuals for foreleg, midleg, hindleg, linearized tympanal size and eye length. Head size was excluded because head length contributed to male morph assignment. The omnibus permutation test detected no difference among correlation matrices ($p = 0.848$).



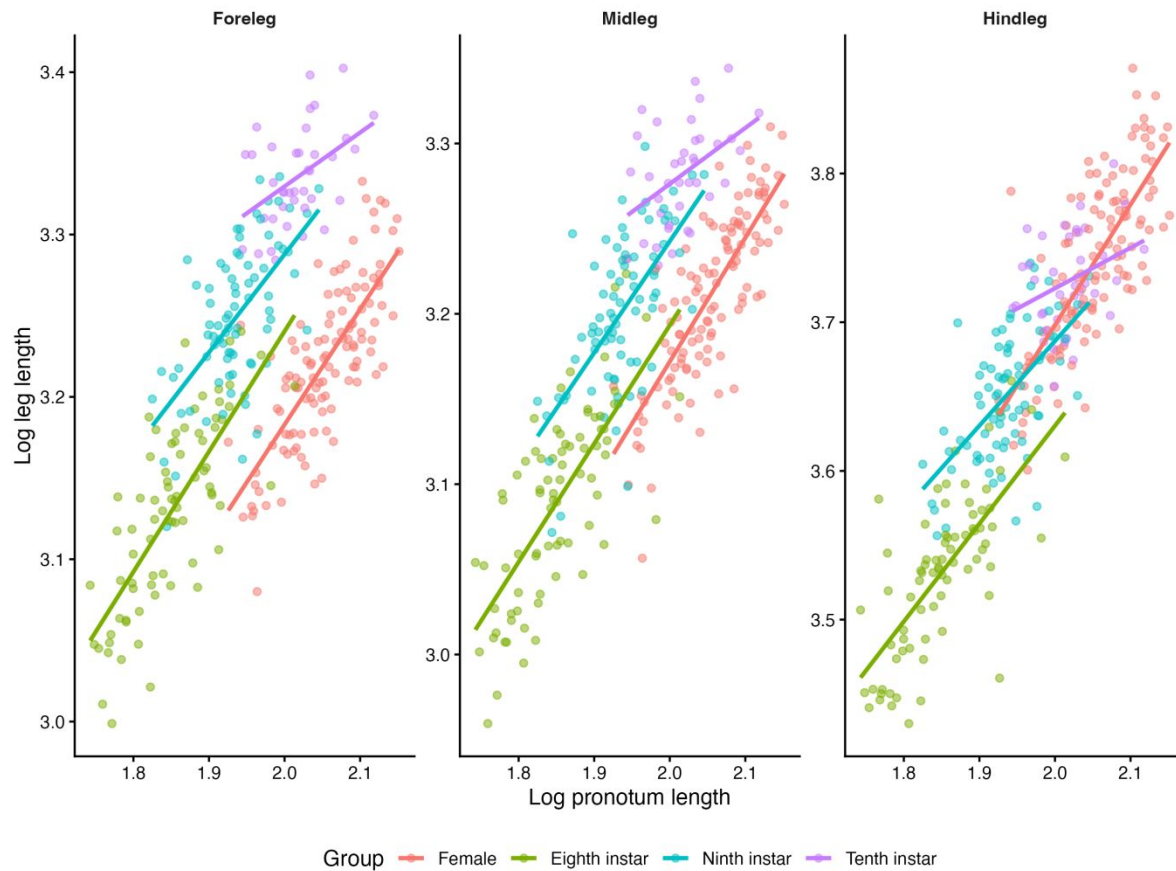
Supplementary Figure S2. Individual scores on the first two principal components of the five nonclassification traits after group-specific adjustment to a pronotum length of 7.2 mm. Diamonds show group centroids and ellipses show descriptive 68% normal regions. Parallel analysis retained PC1 but not PC2.



273 **Supplementary Figure S3. Observed pronotum-length distributions by group. The dashed**
274 **line marks the 7.2-mm reference used for adjusted means and contrasts; it falls within the**
275 **common observed support of all four groups.**



278 **Supplementary Figure S4. Log–log relationships between pronotum length and total**
 279 **foreleg, midleg and hindleg length in females and the three male morphs. Points show**
 280 **individual measurements; fitted lines describe group-specific body-size relationships.**



281

282

Supplementary references

Kelly, C.D. 2026. [Can the form of sexual selection explain patterns of static weapon allometry expressed by alternative mating morphotypes?](#) *Evolution* **80**: 191–201.

Van Noordwijk, A.J. & de Jong, G. 1986. [Acquisition and allocation of resources: Their influence on variation in life history tactics.](#) *Am. Nat.* **128**: 137–142. University of Chicago Press.